# Math 2551 Worksheet: Change of Variables

1. Find the Jacobian determinant of the transformation $x = e^{-r}\sin(\theta), y = e^{r}\cos(\theta)$.

2. Find the image of the set $S$ which is the disk given by $u^2+v^2 \leq 1$ under the transformation

$$\mathbf{T}(u,v) = \begin{bmatrix} au \\ bv \end{bmatrix}$$

   for some constants $a, b > 0$.

   Give an example integration problem for which this result would be helpful.

3. Find equations for a transformation $\mathbf{T}$ that maps a rectangular region $S$ in the $uv$-plane whose sides are parallel to the $u$- and $v$-axes onto the region $R$ bounded by the hyperbolas $y = 1/x, y = 3/x$ and the lines $y = x, y = 3x$ in the first quadrant.

4. Solve the system

$$u = 2x - 3y, v = -x + y$$

   for $x$ and $y$ in terms of $u$ and $v$. Then find the value of the Jacobian and find the image of the parallelogram $R$ in the $xy$-plane with boundaries $x = -3, x = 0, y = x$, and $y = x + 1$ under this transformation. Sketch the transformed region in the $uv$-plane. Use your results to rewrite the integral

$$\iint_R 2(x - y)\ dx\ dy$$

   as an integral in $uv$-coordinates.

5. Use a change of variables to compute

$$\iint_R xy\ dA,$$

   where $R$ is the region in the first quadrant bounded by the lines $y = x$ and $y = 3x$ and the hyperbolas $xy = 1, xy = 3$.

   Hint: Think about problem 3 above.

6. Compute $\displaystyle\iint_R e^{x+y}\ dA$, where $R$ is the region given by the inequality $|x| + |y| \leq 1$.